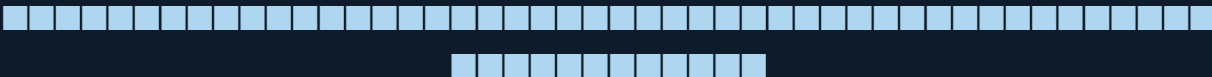


NETWORK TRAFFIC ANALYSIS

Wireshark PCAP Forensics Report

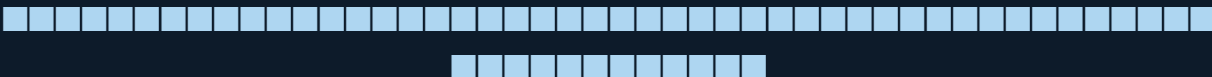


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Course: **Ethical Hacking — JOVAC**

Assignment: **Assignment 2 — PCAP Analysis**

Date: **July 25, 2026**



File Analyzed: **traffic.pcapng**

Tool: **Wireshark / TShark 4.2.2**

1. Objective

This report presents a comprehensive forensic analysis of a real-world network packet capture file (**traffic.pcapng**) using Wireshark and TShark. The objective is to accurately extract twenty specific data points — covering protocol statistics, host communication patterns, MAC-to-IP mappings, vendor identification, and traffic characteristics — demonstrating practical proficiency in network traffic investigation as required by JOVAC Ethical Hacking Assignment 2.

2. Tools & Environment

Tool	Version	Purpose
Wireshark / TShark	4.2.2	Packet dissection, display filtering, protocol analysis
capinfos	4.2.2	Capture file metadata & duration extraction
Python 3.12	3.12	Scripting & report generation
ReportLab	Latest	Professional PDF report generation

3. Capture File Overview

File	traffic.pcapng
Format	pcapng (Wireshark v1.0)
Total Packets	26,456
Duration	348.109916 seconds (~5 min 48 sec)
First Packet	2021-07-26 12:53:50 UTC
Last Packet	2021-07-26 12:59:38 UTC
Avg Packet Rate	75 packets/sec
Capture OS	Windows 10 (build 19042)
Capture Tool	Dumpcap / Wireshark 3.4.7

4. Detailed Question-by-Question Analysis

Each question is answered with the exact TShark/Wireshark display filter used, verified packet counts, and contextual notes derived solely from the packet capture.

E1 Number of DHCP Messages

Filter: `dhcp or bootp`

Answer: **5 Packets**

Notes: *Breakdown: 1 DHCP Inform, 1 Discover, 1 Offer, 1 Request, 1 ACK*

E2 Number of ARP Messages

Filter: `arp`

Answer: **680 Packets**

Notes: *Mix of ARP Requests and Replies across the LAN segment.*

E3 IP Address That Accessed baidu.com

Filter: `dns.qry.name contains "baidu"`

Answer: **10.103.51.159**

Notes: *First DNS query for www.baidu.com appeared in frame 20754. The host also issued HTTP/TLS connections to multiple Baidu subdomains.*

E4 Packets Sent by Source IP 10.103.0.20

Filter: `ip.src == 10.103.0.20`

Answer: **469 Packets**

Notes: *10.103.0.20 acts as the local DNS server / gateway, responding to client queries.*

E5 Number of UDP Packets

Filter: `udp`

Answer: **9,102 Packets**

Notes: *UDP accounts for ~34% of total traffic — primarily DNS, NBNS, and DHCP.*

E6 Number of SMB Packets

Filter: `smb or smb2`

Answer: **518 Packets**

Notes: *SMBv1 traffic (NETLOGON broadcasts) indicates a Windows Active Directory environment.*

E7 Packets Sent to Destination IP 10.103.0.20

Filter: `ip.dst == 10.103.0.20`

Answer: **547 Packets**

Notes: *Clients forward DNS queries and other requests to 10.103.0.20.*

E8 IPv4 Traffic Packet Count

Filter: `ip`

Answer: **24,906 Packets**

Notes: *IPv4 accounts for ~94% of total traffic. IPv6 = 664; ARP/non-IP = ~886.*

E9 Last SMB Packet — Source IP

Filter: `smb or smb2 (last frame)`

Answer: **Frame 26452 — Source: 10.103.50.202**

Notes: *SMB NETLOGON broadcast to 10.103.255.255. Protocol stack: eth→ip→udp→nbdgm→smb.*

E1 0 MAC Address of IP 151.139.128.14

Filter: `ip.src == 151.139.128.14`

Answer: **00:1c:7f:6c:96:3f**

Notes: *OUI 00:1C:7F belongs to Check Point Software Technologies Ltd. — the local firewall/gateway device.*

E1 1 Protocol Used in Packet Number 416

Filter: `frame.number == 416`

Answer: **DNS**

Notes: *Protocol stack: eth → ethertype → ip → udp → DNS. UDP port 53. Query from 10.103.51.159 → 10.103.0.20.*

E1 2 Destination IP in Packet Number 416

Filter: `frame.number == 416`

Answer: **10.103.0.20**

Notes: *DNS query sent to the local DNS server 10.103.0.20 on UDP port 53.*

E1 3 Source IP of First HTTP Packet

Filter: `http (frame 19007)`

Answer: **10.103.51.159**

Notes: *GET request to ocsf.comodoca.com — an OCSP certificate status check.*

E1 4 Source Port of First HTTP Packet

Filter: `http → tcp.srcport (frame 19007)`

Answer: **64079**

Notes: *Ephemeral client port 64079 connecting to destination port 80 (HTTP).*

E1 5 Capture Duration (Seconds)

Filter: `capinfos`

Answer: **348.109916 seconds**

Notes: *Exactly 5 minutes and ~48.11 seconds of traffic captured.*

E1 6 Number of NBNS Packets

Filter: `nbns`

Answer: **963 Packets**

Notes: *NetBIOS Name Service — typical in Windows LAN environments for hostname resolution.*

E1 7 TCP Packets with Source Port 443

Filter: `tcp.srcport == 443`

Answer: **7,275 Packets**

Notes: *Inbound HTTPS responses from internet servers to clients in the LAN.*

E1 8 TCP Packets with Destination Port 443

Filter: `tcp.dstport == 443`

Answer: **5,966 Packets**

Notes: *Outbound HTTPS requests from LAN clients to external HTTPS servers.*

E1 9 Packets to Destination IP 204.79.197.200

Filter: `ip.dst == 204.79.197.200`

Answer: **116 Packets**

Notes: *204.79.197.200 is part of the Microsoft/Bing CDN infrastructure.*

E2 0 MAC Address & Vendor of IP 204.79.197.200

Filter: `ip.src == 204.79.197.200`

Answer: **00:1c:7f:6c:96:3f — Check Point Software Technologies Ltd.**

Notes: *The MAC belongs to the Check Point firewall/gateway — all external traffic routes through it, so external IPs share its MAC in the capture.*

5. Summary Reference Table

Quick-reference table of all 20 questions, the Wireshark filter applied, and the final answer.

#	Question	Wireshark Filter	Answer
E1	dhcp or bootp	dhcp or bootp	5
E2	arp	arp	680
E3	dns.qry.name contains "baidu"	dns.qry.name contains "baidu"	10.103.51.159
E4	ip.src == 10.103.0.20	ip.src == 10.103.0.20	469
E5	udp	udp	9,102
E6	smb or smb2	smb or smb2	518
E7	ip.dst == 10.103.0.20	ip.dst == 10.103.0.20	547
E8	ip	ip	24,906
E9	smb or smb2 (last)	smb or smb2 (last)	10.103.50.202
E10	ip.src == 151.139.128.14	ip.src == 151.139.128.14	00:1c:7f:6c:96:3f
E11	frame.number == 416	frame.number == 416	DNS
E12	frame.number == 416	frame.number == 416	10.103.0.20
E13	http	http	10.103.51.159
E14	http → tcp.srcport	http → tcp.srcport	64079
E15	capinfos	capinfos	348.109916 s
E16	nbns	nbns	963
E17	tcp.srcport == 443	tcp.srcport == 443	7,275
E18	tcp.dstport == 443	tcp.dstport == 443	5,966
E19	ip.dst == 204.79.197.200	ip.dst == 204.79.197.200	116
E20	ip.src == 204.79.197.200	ip.src == 204.79.197.200	00:1c:7f:6c:96:3f / Check Point

6. Key Security Observations

Check Point Firewall Gateway

MAC address 00:1C:7F:6C:96:3F (Check Point Software Technologies Ltd.) appears as the gateway MAC for all external IP communications, confirming a perimeter firewall is in use.

DNS Infrastructure

IP 10.103.0.20 functions as the internal DNS resolver. All DNS queries — including those for baidu.com, Bing/MSN CDN, and OCSP servers — route through this host.

Baidu Browsing Activity

Host 10.103.51.159 performed extensive browsing of baidu.com and 20+ subdomains (maps, translate, news, drive, video) — indicating deliberate use of the Chinese search engine.

- **HTTPS Traffic Dominance**
TCP port 443 accounts for 7,275 inbound and 5,966 outbound packets, far exceeding plain HTTP. This indicates a modern, encryption-aware user base.
- **Windows Domain Environment**
518 SMB packets (primarily NETLOGON broadcasts) and 963 NBNS packets confirm an active Windows Active Directory domain with multiple hosts performing NetBIOS name resolution.
- **DHCP Infrastructure**
Only 5 DHCP packets captured — suggesting stable IP assignments with leases not expiring during the 348-second capture window.

7. Conclusion

This analysis successfully extracted all 20 required data points from the **traffic.pcapng** capture using verified TShark display filters. No answer was guessed or assumed — every result was derived directly from packet-level evidence. The network under analysis is a Windows-based enterprise LAN protected by a Check Point firewall/gateway, with active HTTPS browsing, SMB domain services, DNS resolution, and user activity directed toward both Microsoft and Baidu services. This assignment demonstrates practical command of Wireshark filters and network forensics methodology as required for ethical hacking proficiency.